Cheat sheet

# Git, Linux & Docker

The developer toolkit every AI engineer uses

## Git

terminal

```
git clone URL; git status; git add -A; git commit -m "msg"; git push
git switch -c feature/rag; git merge main; git rebase main
git log --oneline --graph; git diff; git stash; git restore file
git revert <sha>        # safe undo on shared branches
```

| | |
|---|---|
| **Branching** | Feature branches + pull requests + code review |
| **.gitignore** | `.env`, `venv/`, `__pycache__/`, `*.pkl`, `node_modules/` |
| **Merge vs rebase** | Merge keeps history; rebase rewrites for a linear history (never on shared branches) |

## Linux

terminal

```
ls -la; cd; pwd; mkdir -p a/b; cp -r; mv; rm -r; cat; less; head; tail -f log.txt
grep -rn "error" .; find . -name "*.py"; chmod +x run.sh; ps aux; kill PID; top
curl -s URL | jq '.data'; export KEY=value; echo $PATH; ssh user@host
```

## Docker

terminal

```
docker build -t app .; docker run -p 8000:8000 --env-file .env app
docker ps; docker logs -f ID; docker exec -it ID sh; docker compose up -d
```

| | |
|---|---|
| **Image vs container** | Blueprint vs running instance |
| **Layers and cache** | Copy requirements and install before copying code |